

ARAVINDA APARNA VELUMURI

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PROFESSIONAL EXPERIENCE

Data Engineer

Jan 2025 - Aug 2025

Skillz Inc

Las Vegas, NV

- Developed scalable Reverse ETL pipeline enabling real-time data syncs for personalized campaigns, reducing manual analytics queries by 75% and delivering \$54K/year in cost savings.
- Collaborated with product, finance, and analytics teams to build an end-to-end ad-revenue system supporting 40K+ daily impressions and driving \$500K+ annual uplift from non-cash users.
- Boosted ROAS by 53% and improved user retention by integrating ad-platform APIs via Airflow DAGs to enhance event attribution accuracy.
- Streamlined HR data ingestion into Snowflake, powering Tableau dashboards and eliminating 12+ hours/week of manual reporting.
- Migrated AWS DMS replication and deployed Kinesis and Snowpipe jobs using Terraform, improving production reliability and reducing latency by 25%.
- Implemented and maintained real-time segmentation (RTS) pipelines using Apache Flink on AWS, integrated with Amazon MSK and Kafka Connect, ensuring reliable streaming of user segmentation data to downstream systems.

Software Engineer

Oct 2022 - May 2023

Audintel Inc

Hyderabad, India

- Designed and developed responsive React.js components for login, account, and profile modules, contributing to 47% increase in user engagement through improved UI/UX.
- Reduced API response time by 23% by designing and implementing RESTful APIs in Node.js/Express for authentication and low-latency client-server communication.
- Optimized SQL queries and indexing strategies, improving data retrieval performance by 18% for high-traffic endpoints, leveraging database design principles.

Data Science Intern

May 2022 - July 2022

International Institute of Information Technology

Hyderabad, India

- Designed ETL pipelines to collect and process 100K+ school profiles from DBpedia and Wikipedia using Selenium, BeautifulSoup, and Python, reducing manual intervention by 12%.
- Improved data reliability and processing efficiency by 15% by engineering workflows for data wrangling, cleaning, normalization, and deduplication of structured and semi-structured datasets.
- Automated data extraction and documentation generation for academic research project, processing 100K+ educational profiles using Jinja2 templates and XML transformation.

PROJECTS

Real-Time Stock Market Streaming Pipeline

Sep 2024 - Dec 2024

- Developed a real-time data pipeline using Apache Kafka and Spark Structured Streaming to ingest and process live stock market feeds, processing 10K+ events/sec with sub-second latency.
- Implemented watermarking, windowing, and checkpointing to S3/HDFS for fault tolerance and accurate handling of late-arriving events, and visualized processed insights on a Streamlit dashboard.

Multimodal RAG for Media Content Intelligence

Sep 2025 - Dec 2025

- Architected multimodal RAG pipeline processing images, audio, and PDFs using CLIP for visual embeddings, Whisper for transcription, and ChromaDB vector storage, indexing 1,000+ media assets with less than 300ms query latency.
- Engineered cross-modal semantic search enabling natural language queries across multiple content types using Qwen-VL for response generation.

TECHNICAL SKILLS

Programming: Python, SQL, Java, JavaScript, R

Data Engineering: Airflow, Snowflake, Databricks, Kafka, Spark, Flink, Docker, Kubernetes, dbt, PySpark

Databases: MySQL, Redshift, Redis, DynamoDB, MongoDB, ChromaDB, Pinecone

Tools: AWS, Git (CI/CD), Terraform, Tableau, Power BI, Datadog, Jenkins, Argo CD, Jira, Excel

EDUCATION

Northeastern University, Boston, MA

Sep 2023 - Dec 2025

Master of Science in Data Science

GPA: 3.9/4.0

Courses: Database Management Systems, Machine Learning, LLM, Operating Systems, Cloud, Algorithms, Web Development

Gokaraju Rangaraju Institute of Engineering and Technology, India

Aug 2019 - Jun 2023

Bachelor of Technology - Information Technology

GPA: 3.6/4.0

LEADERSHIP AND ACTIVITIES

- Mentored 300+ students as a **Teaching Assistant** for SQL and Python Programming courses, teaching query optimization, joins, indexing strategies, database normalization, Python-based EDA and feature engineering, data visualization, and ETL concepts.
- Collegiate table tennis runner-up, Las Vegas Corporate Challenge semi-finalist (2025), and college art & craft competition winner.
- Organized technical hackathons, workshops, and project expos, coordinating with faculty, students, and industry mentors.